TRIDIAGONAL DETERMINANTS AND THE FIBONACCI MATRIX

From cubes to Fibonacci

Change the entries, grow the matrix, and discover an addition rule.

1 / PICK UP THE CUBIC EXAMPLE

Set $c = 0$, then change b

The cyclic matrix from the previous page becomes

$$C_3(a,b) = \begin{pmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{pmatrix}, \qquad \det C_3 = a^3 + b^3.$$

The power of b decides what a substitution does:

Replace b by	New determinant
b	$a^3 + b^3$
$-b$	$a^3 - b^3$
$\mathrm{i}b$	$a^3 - \mathrm{i}b^3$

$$\mathrm{i}^2 = -1, \qquad \mathrm{i}^3 = -\mathrm{i}.$$

Gold marks the closing link. Next, use both neighbouring diagonals and zero far corners.

2 / KEEP ONLY THE THREE CENTRAL DIAGONALS

A tridiagonal pattern

Only the main diagonal and its two neighbours can have non-zero entries:

$$T_3(a,b) = \begin{pmatrix} a & b & 0 \\ b & a & b \\ 0 & b & a \end{pmatrix}.$$

Write $D_n = \det T_n(a,b)$ as this pattern grows. Use $D_0 = 1$, $D_1 = a$. The first cases are

$$D_2 = a^2 - b^2,$$
$$D_3 = a(a^2 - b^2) - b(ab).$$

Expanding along the first row gives the same rule at every size:

$$\boxed{D_n = aD_{n-1} - b^2 D_{n-2}} \quad (n \geq 2).$$

Here $b \mapsto -b$ leaves D_n unchanged.

3 / LET AN IMAGINARY ENTRY CHANGE THE RULE

A minus becomes a plus

Choose $a = 1$ and $b = \mathrm{i}$. For example,

$$T_4(1,\mathrm{i}) = \begin{pmatrix} 1 & \mathrm{i} & 0 & 0 \\ \mathrm{i} & 1 & \mathrm{i} & 0 \\ 0 & \mathrm{i} & 1 & \mathrm{i} \\ 0 & 0 & \mathrm{i} & 1 \end{pmatrix}.$$

Since $b^2 = \mathrm{i}^2 = -1$, the determinant rule becomes

$$D_n = D_{n-1} - (-1)D_{n-2}.$$

$$\boxed{D_n = D_{n-1} + D_{n-2}}.$$

Start with $D_0 = 1$, $D_1 = 1$, and add the last two:

n	0	1	2	3	4	5
D_n	1	1	2	3	5	8

With $F_0 = 0$, $F_1 = 1$, these are Fibonacci numbers shifted one place: $D_n = F_{n+1}$.

4 / STORE TWO SUCCESSIVE VALUES

The same addition in a 2×2 matrix

Let

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Multiplication does two things: add the entries, then keep the previous first entry:

$$Q \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u+v \\ u \end{pmatrix}.$$

Starting with $\binom{1}{1}$, repeated multiplication gives

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{Q} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \xrightarrow{Q} \begin{pmatrix} 3 \\ 2 \end{pmatrix} \xrightarrow{Q} \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The same values appear inside the powers of Q:

$$Q^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad Q^3 = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}.$$

$$\begin{pmatrix} D_n \\ D_{n-1} \end{pmatrix} = Q \begin{pmatrix} D_{n-1} \\ D_{n-2} \end{pmatrix} \quad (n \geq 2).$$

The link: one recurrence, two matrix constructions

$$\det T_n(1,\mathrm{i}) = F_{n+1} \qquad Q^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \quad (n \geq 1).$$

Companion to the difference-of-squares page. $D_0 = 1$ is the empty-determinant convention. The cyclic matrix C_3 and the tridiagonal family T_n use different entry patterns.

Further reading. Cahill and Narayan, *Fibonacci and Lucas Numbers as Tridiagonal Matrix Determinants*; H. W. Gould, the Fibonacci Q-matrix.

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